Corporate Finance

Lecture 1: Capital Markets and the Pricing of Risk

Camilo Saldarriaga

Finance Department, ESCP Business School, 2026/2027

Plan

- Learn some general concepts
- Learn how to compute returns
- Understand how we measure risk
- Understand what is an index

Key concepts

- Compute/estimate returns and risk
- Risk-return relationship of assets

Outline

A. Risk and return: a first glimpse (Chap. 10.1)

B. Common return measures (Chap. 10.2)

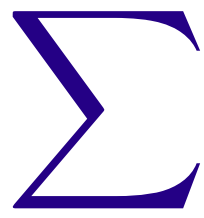
C. Historical returns of stocks and bonds (Chap. 10.3)

D. Common measures of risk (Chap. 10.2)

E. The historical risk-return trade-off (Chap. 10.4)

F. Risk and returns of individual stocks (Chap. 10.4)

But first, let's battle a monster.



This is a monster you will find along the course several times

Initially it may look scary...



But hopefully, you will see that the monster is not that scary



The monster only means “add these up”

- The symbol $\sum$ is the Greek letter S, and S stands for **sum**. It is a shortcut for writing a long addition.
- Instead of writing the addition in full, we can write it with the symbol:

$$1 + 2 + 3 + 4 + 5 \qquad \text{is written} \qquad \sum_{i=1}^{5} i$$

- Both mean exactly the same thing: **15**. The symbol saves ink, nothing more.

How to read the monster

- Three things are attached to it:
  - Below: where to **start** counting, here $i = 1$.
  - Above: where to **stop**, here 5.
  - To the right: **what to add**, here just i itself.
- So you take the recipe on the right, put in $i = 1$, then $i = 2$, and so on until you reach the number on top, and add up everything you wrote:

$$\sum_{i=1}^{5} i = \underbrace{1}_{i=1} + \underbrace{2}_{i=2} + \underbrace{3}_{i=3} + \underbrace{4}_{i=4} + \underbrace{5}_{i=5} = 15$$

The recipe can be anything

- The letter i is just a counter. Whatever sits to the right of the symbol is the recipe, and the counter is replaced by 1, then 2, then 3...
- Adding the squares of the first four numbers:

$$\sum_{i=1}^{4} i^2 = 1^2 + 2^2 + 3^2 + 4^2 = 1 + 4 + 9 + 16 = 30$$

- In finance the recipe is usually a cash flow to be discounted:

$$\sum_{i=1}^{3} \frac{C_i}{(1+r)^i} = \frac{C_1}{1+r} + \frac{C_2}{(1+r)^2} + \frac{C_3}{(1+r)^3}$$

Summing up to n

- Often we do not know how many terms there are, so we call that number n:

$$\sum_{i=1}^{n} \frac{C_i}{(1+r)^i} = \frac{C_1}{1+r} + \frac{C_2}{(1+r)^2} + \cdots + \frac{C_n}{(1+r)^n}$$

- It says: start at the first year, stop at year n, and add up every discounted cash flow in between.
- A project paying for 5 years has $n = 5$; one paying for 30 years has $n = 30$. The formula does not change, only where we stop.

Summing up to infinity

- Sometimes the payments never stop, for instance a company that is expected to pay a dividend every year forever. Then we write ∞ on top:

$$\sum_{i=1}^{\infty} \frac{C}{(1+r)^i} = \frac{C}{1+r} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \ldots$$

- Adding infinitely many numbers does not have to give an infinite total. Each term here is smaller than the one before, because we divide by $(1+r)$ once more every year.
- The terms shrink fast enough that the total settles on a finite number. This is the perpetuity, and it is why a share paying dividends forever still has a finite price.

Why the total can be finite

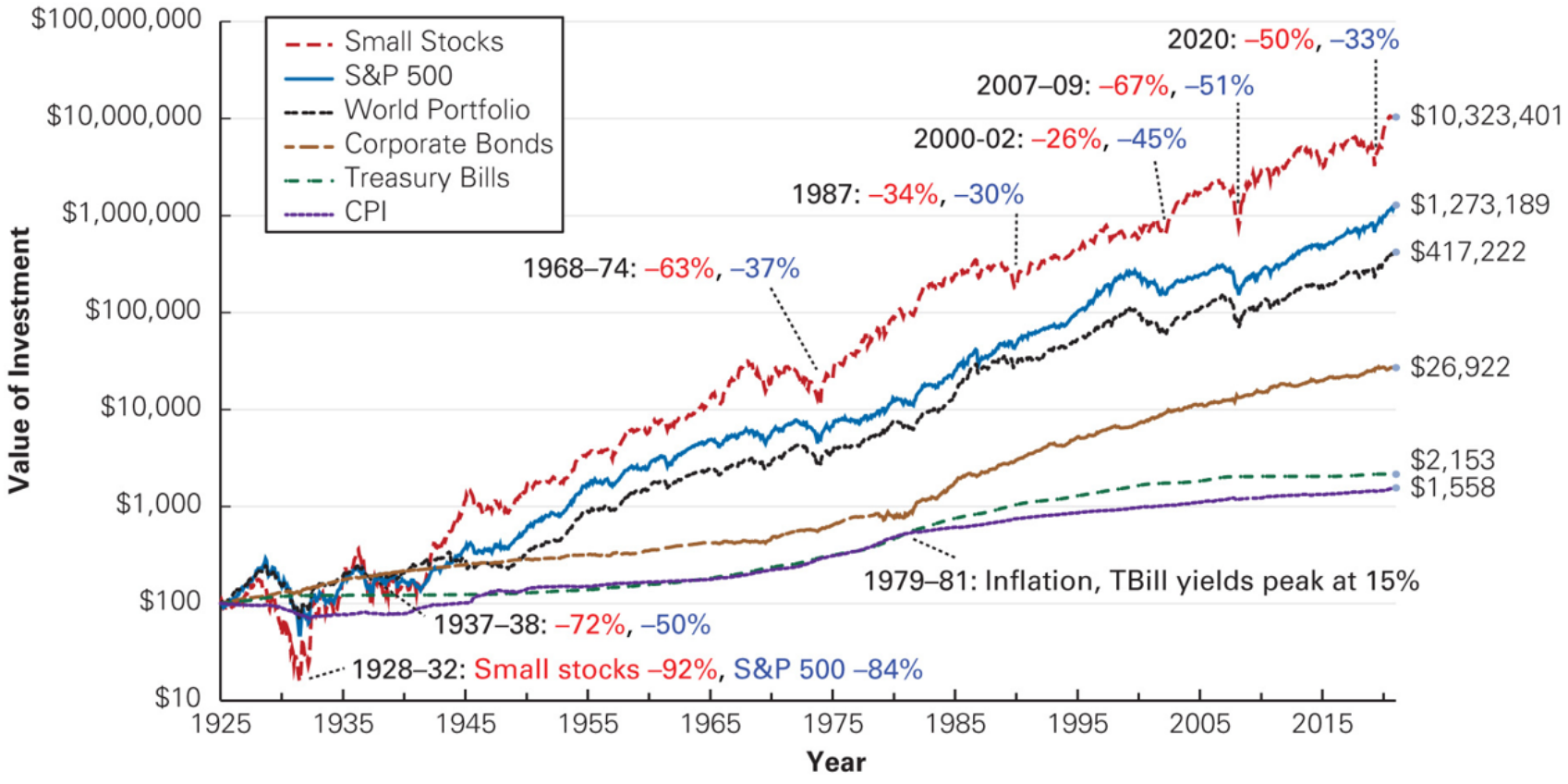
- A simpler example of the same idea: keep adding halves of what is left:
$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \ldots$$
- After two terms you are at 0.75, after three at 0.875, after four at 0.9375. The total never passes 1, it just gets closer and closer.
- The monster is tamed: an endless sum of shrinking terms adds up to a number you can actually put in a valuation.

A. Risk and return: a first glimpse

- Standard & Poor's 500: 90 U.S. stocks up to 1957 and 500 after that. Leaders in their industries and among the largest firms traded on U.S. Markets.
- Small stocks: securities traded on the NYSE with market capitalizations in the bottom 20%.
- MSCI World Index (World portfolio)
- Corporate Bonds: long-term, AAA-rated U.S. corporate bonds with maturities of approximately 20 years.
- Treasury Bills: an investment in three-month Treasury bills.
- CPI (consumer price index)

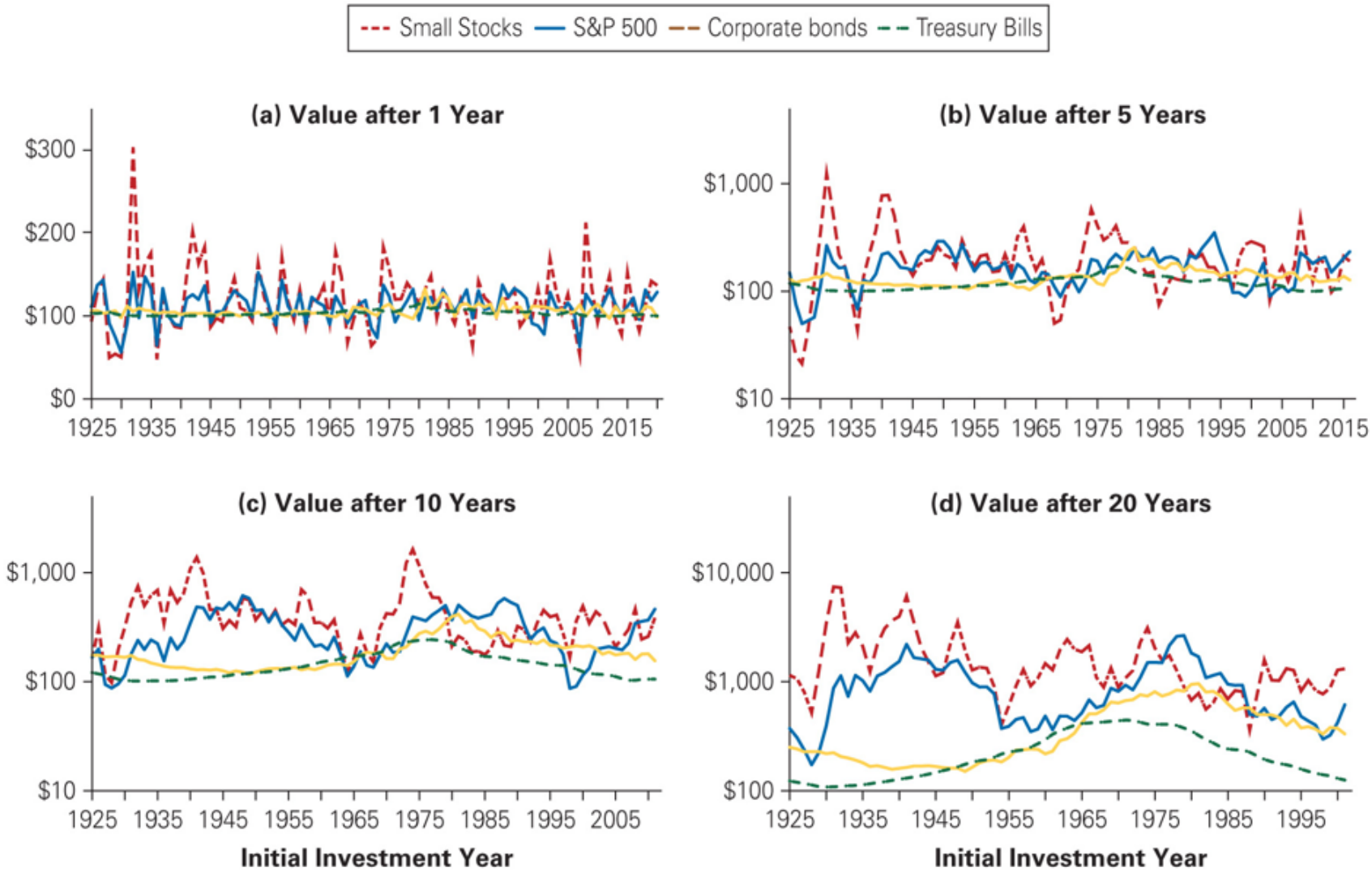
$\Rightarrow$ We ignore transaction costs and assume that interest and dividends are reinvested.

Value of $100 invested at the end of 1925



Source: CRSP, Standard and Poor's, MSCI, Global Financial Data.

Value of $100 invested in different years and for differing horizons



Source: CRSP, Standard and Poor's, MSCI, Global Financial Data.

B. Common return measures

How to determine the return of a financial asset?

- Returns from dividends and capital gains
- Calculations with reinvested dividends
- Stock market indices
- Histograms of historical returns
- Historical volatility
- Arithmetic mean versus geometric mean

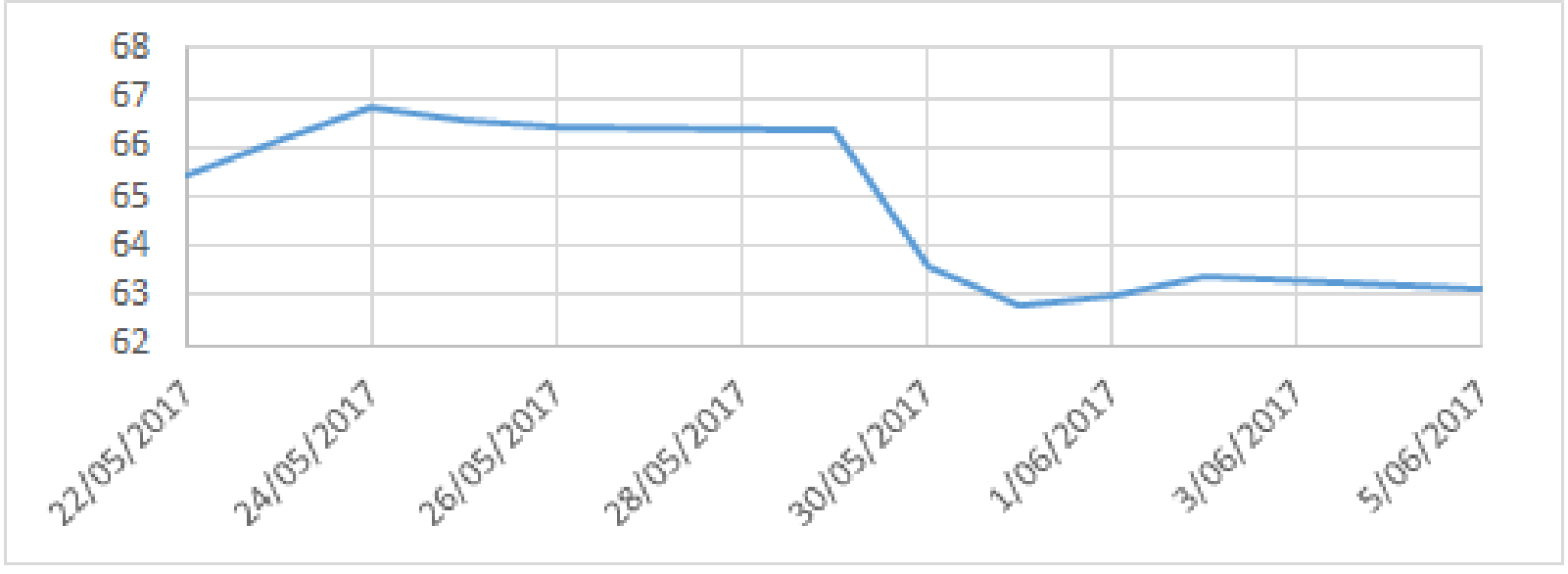
Dividend return and capital gains

- The realized return of a stock between two dates consists of:
  - capital gains and
  - dividends

$$
\begin{aligned}
R_{t+1} &= \frac{\text{Wealth at period end}}{\text{Initial investment}} - 1 \\
&= \frac{P_{t+1} + Div_{t+1} - P_t}{P_t} \\
&= \underbrace{\frac{P_{t+1} - P_t}{P_t}}_{\text{Capital Gain Rate}} + \underbrace{\frac{Div_{t+1}}{P_t}}_{\text{Dividend Yield}}
\end{aligned}
$$

Stock prices and dividend payment: the example of BNP Paribas

A dividend of EUR 2.7 is detached from the share price on May 30th, 2017



Source: Yahoo Finance

How to calculate the average return on an investment?

- What is the better investment (Investment at the end of year N until year N+2)?

Investment 1				
Value	100	50	100	Average
Return		-50%	+100%	+25 %
Investment 2				
Value	100	110	121	
Return		+10%	+10%	+10%

- The arithmetic average is misleading ...

Geometric mean (annualized), or CAGR

- CAGR stands for **Compound Annual Growth Rate**: the constant yearly return that turns the initial investment into the final wealth.
- The geometric mean is calculated as

$$\text{Geometric mean} = \left(\frac{\text{Wealth at the end of n years}}{\text{Initial investment}}\right)^{\frac{1}{n}} - 1$$

- or

$$\text{Geometric mean} = \left(\left(1 + R_1\right)\left(1 + R_2\right) \ldots \left(1 + R_n\right)\right)^{\frac{1}{n}} - 1$$

To sum up: calculation of returns

- Depending on the situation, it is best to calculate mean returns differently. In general:
  - Intertemporal average or realized returns: geometric mean (=IRR of the investment, or "compound annual return")
  - Mean return of multiple investments or expected return: arithmetic mean
- The two means are equal if annual returns are constant. Otherwise the arithmetic mean is always larger than the geometric mean.
- The geometric mean is approximately equal to the arithmetic mean - $\frac{1}{2}$ variance.

Compounding returns

- The returns over multiple periods need to be compounded to obtain the return over a longer horizon.
- Example: Over 4 consecutive periods an investment generates the returns: 6.1%, -5.32%, 6.72% and 8.93%. The overall return is:

$$\begin{aligned} R_{1\to 4} &= (1+R_1)(1+R_2)(1+R_3)(1+R_4) - 1 \\ &= (1+0.061)(1-0.0532)(1+0.0672)(1+0.0893) - 1 \\ &= 16.7\% \end{aligned}$$

- Remarks:
  - You must not take the sum of the returns!
  - The length of the periods does not matter here.

Example: accounting for reinvested dividends to compute the returns on Total

What was the total return of the Total stock in 2016 (rounded to 4 decimal points)?

Date	Price (in EUR)	Dividend (in EUR)	Return
01/01/2016	40.89		
21/03/2016	41.33	0.61	2.57%
06/06/2016	42.47	0.61	4.23%
27/09/2016	40.77	0.61	-2.57%
21/12/2016	47.60	0.61	18.25%
31/12/2016	48.72		2.35%

Returns and "stock splits"

- Companies sometimes decide to do "stock splits", or "reverse stock splits", which need to be taken into account when returns are calculated.
- Example:
- You bought 100 stocks of CIGNA Corp. on August 5th, 2006 for a price of \$155.59. One year later, the stock is worth \$50.18, but the company has done a "3:1 stock split" so you now own 300 stocks. Over this period, you received a total amount of dividends of \$8.5 (on all your stocks). Your total return is

$$\frac{300 \times 50.18 + 8.5 - 100 \times 155.59}{100 \times 155.59} = -3.2\%.$$

Value of a stock and performance index

- The previous figures don't correspond to the evolution of a specific stock price.
  - They represent the evolution of an investment in the stock.
- They are a "performance index" and/or use "adjusted prices".
- The adjusted stock price accounts for dividends and stock splits. They are calculated such that
  - The last value corresponds to the stock price.
  - The previous values mirror the evolution of an investment in the stock.

Example: the long-term return on a stock

- On January 2nd, 1970, the Coca Cola stock price was worth \$82.25. On June 30th, 2017, the stock price was worth \$44.85. Good or bad investment?
- Coca Cola has done 5 stock splits and made many dividend payments, the "adjusted close" on 2.1.1970 was 0.85.
- Hence, the return on the overall period 1970-2017 is $44.85/0.85 - 1 = 5176\%$.
- The geometric annual mean is $(44.85/0.85)^{1/47} - 1 = 8.80\%$.

Stock price indices

- The stock price indices aggregate the information about the evolution of a stock exchange by depicting the evolution of a portfolio of stocks.
- Example: CAC 40, DAX, S&P 500, FTSE100 (Footsie), Nasdaq 100, Dow Jones Industrial average, Hang Seng, Nikkei 225 etc.
- Some of these indices are very old!
  - Dow Jones has published its index since 1896 in New-York
- **Attention**: These indices can be calculated as "total return index" including dividends or "price index" without dividends! Some indices exist in different versions.

Which indices include dividends?

Gross total return: dividends reinvested	Price only
DAX 40 (Germany)	S&P 500, Dow Jones, Nasdaq (US)
Ibovespa (Brazil)	FTSE 100 (UK)
	Nikkei 225 (Japan)
	CAC 40 (France)
	Hang Seng (Hong Kong)

- **Price index**: when a company pays a dividend, its share price drops, and the index does not add that cash back. It only measures price changes.
- **Total return index**: assumes every dividend is reinvested in the index, before tax, so it grows faster over time.
- Most price indices also have an official total return version, e.g. the S&P 500 TR or the CAC 40 GR.

The same index comes in three versions

- **Price index** (price-only or capital index): follows share prices only. When a company pays a dividend its price drops and the index drops with it; the cash is never added back. The headline S&P 500, Dow Jones, FTSE 100, CAC 40 and Nikkei 225 are all price indices.
- **Gross total return index** (gross return or performance index): every dividend is reinvested in the index on the day it is paid, before any tax. Examples: the DAX 40 as quoted in the news, the S&P 500 TR, the CAC 40 GR.
- **Net total return index** (net return index): dividends are reinvested after the withholding tax a foreign investor would pay. The usual benchmark for international funds and ETFs. Examples: MSCI World Net, S&P 500 Net.
- Same companies, same prices in all three. The only difference is what happens to the dividends, and over the long run that difference is large.

CAC40 and CAC40 GR (gross return)



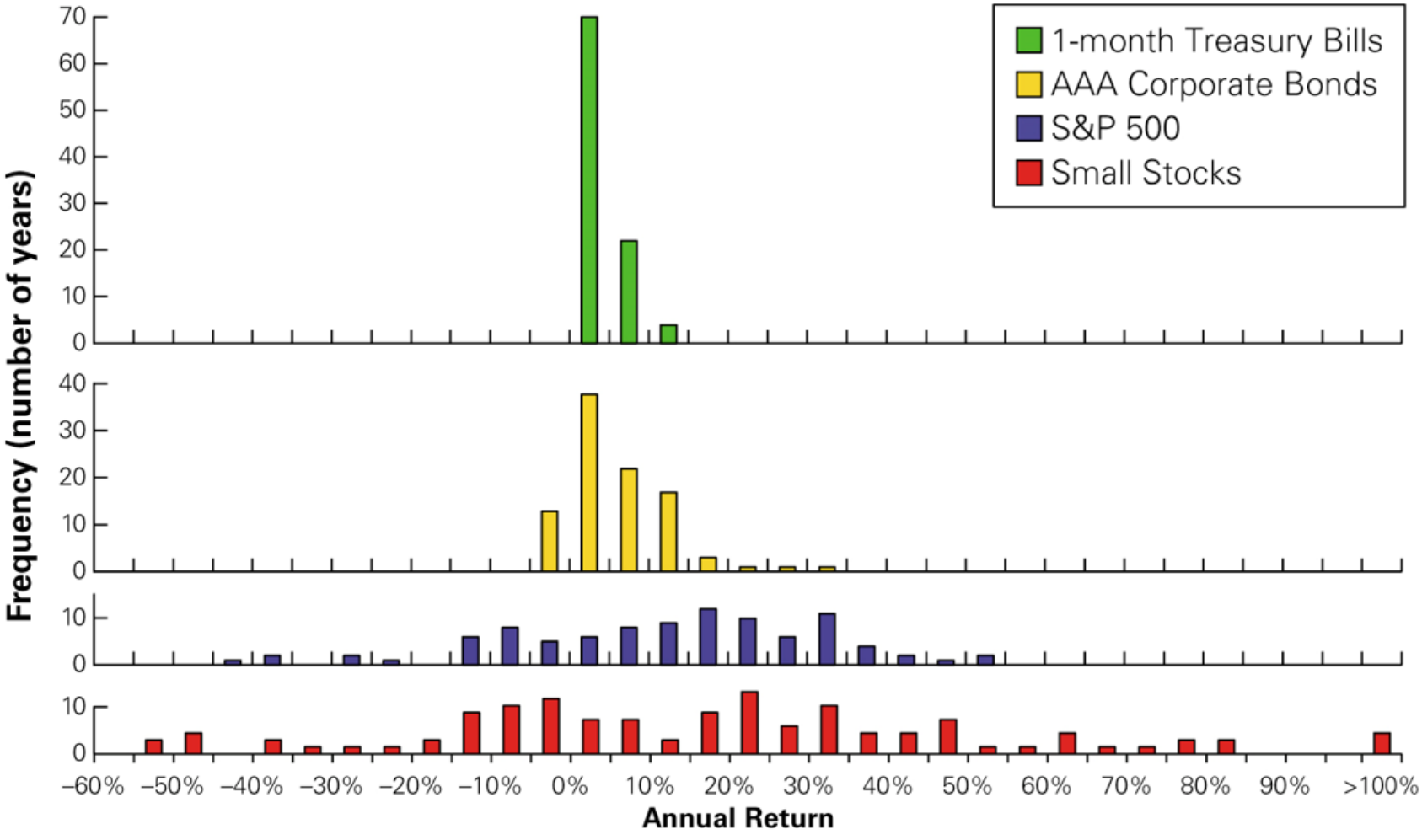
C. Historical returns of stocks and bonds

Annual mean return

$$\overline{R} = \frac{1}{T}\sum_{t=1}^{T} R_t = \frac{1}{T}(R_1 + \cdots + R_T)$$

Investment	Average Annual Return
Small stocks	18.6%
S&P 500	12.2%
Corporate bonds	6.2%
Treasury bills	3.3%

Return histograms of different US portfolios (1926-2021)



D. Common measures of risk

- There is no perfect and generally accepted risk measure.
- In finance, risk is measured by volatility.
- Volatility = standard deviation of the returns

  First calculate the variance, then take its square root to get the volatility:

$$\text{Var}[R] = \frac{1}{T-1}\sum_{t=1}^{T}\left(R_t - \overline{R}\right)^2 \qquad \sigma_R = \sqrt{\text{Var}[R]}$$

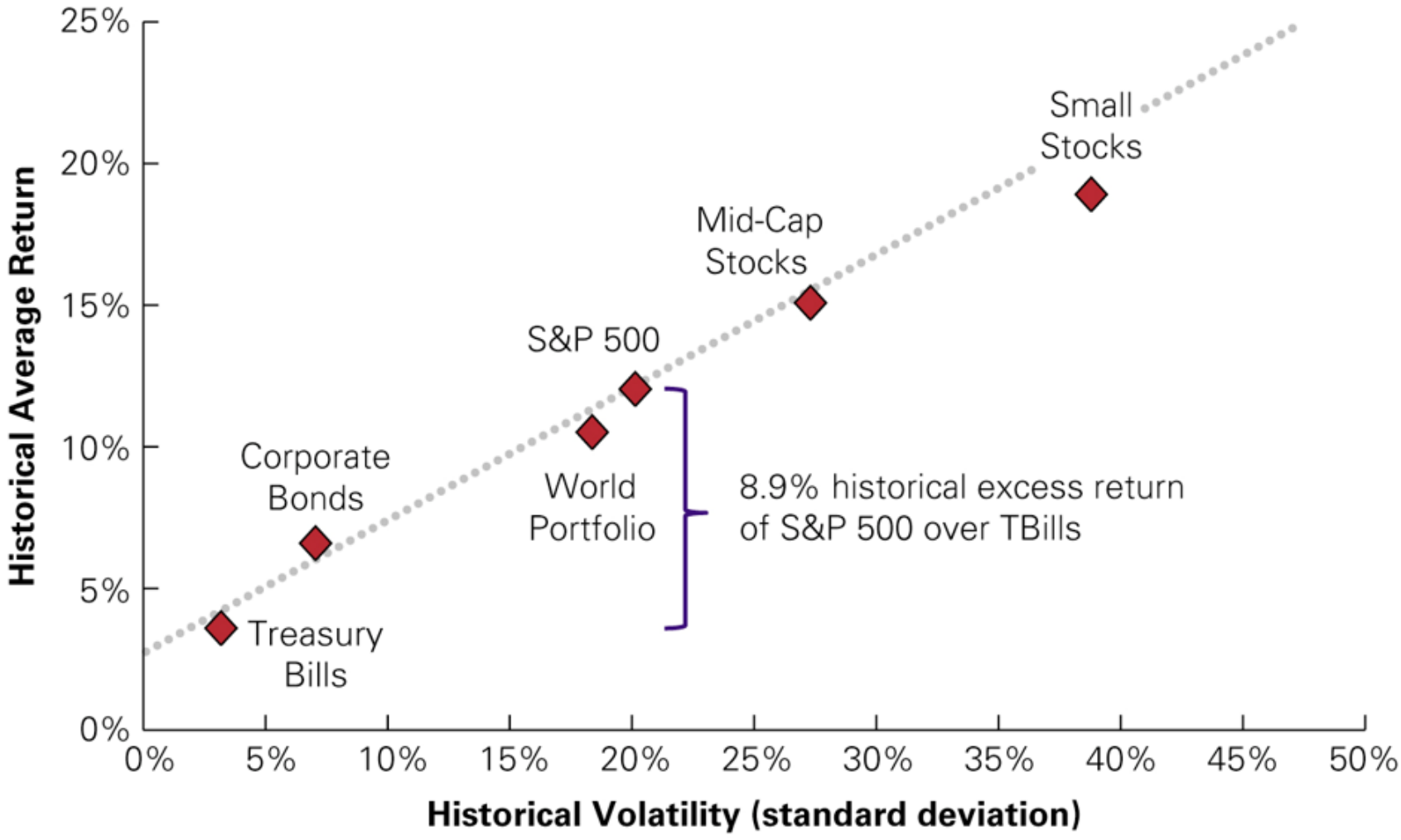
- Attention:
  - The volatility is a symmetric risk measure. It takes into account the risk to win as well as the risk to lose!
  - Other measures capture only the risk of losses (downside risk), e.g. VaR ("value at risk").

E. The historical risk and return trade-off

Investment	Return Volatility (Standard Deviation)	Excess Return (Average Return in Excess of Treasury Bills)
Small stocks	38.6%	15.3%
S&P 500	19.7%	8.9%
Corporate bonds	6.3%	2.9%
Treasury bills (30-day)	3.1%	

$$\begin{aligned}\text{Excess return} &= \text{return in excess of the risk-free rate} \\ &= \text{risk premium}\end{aligned}$$

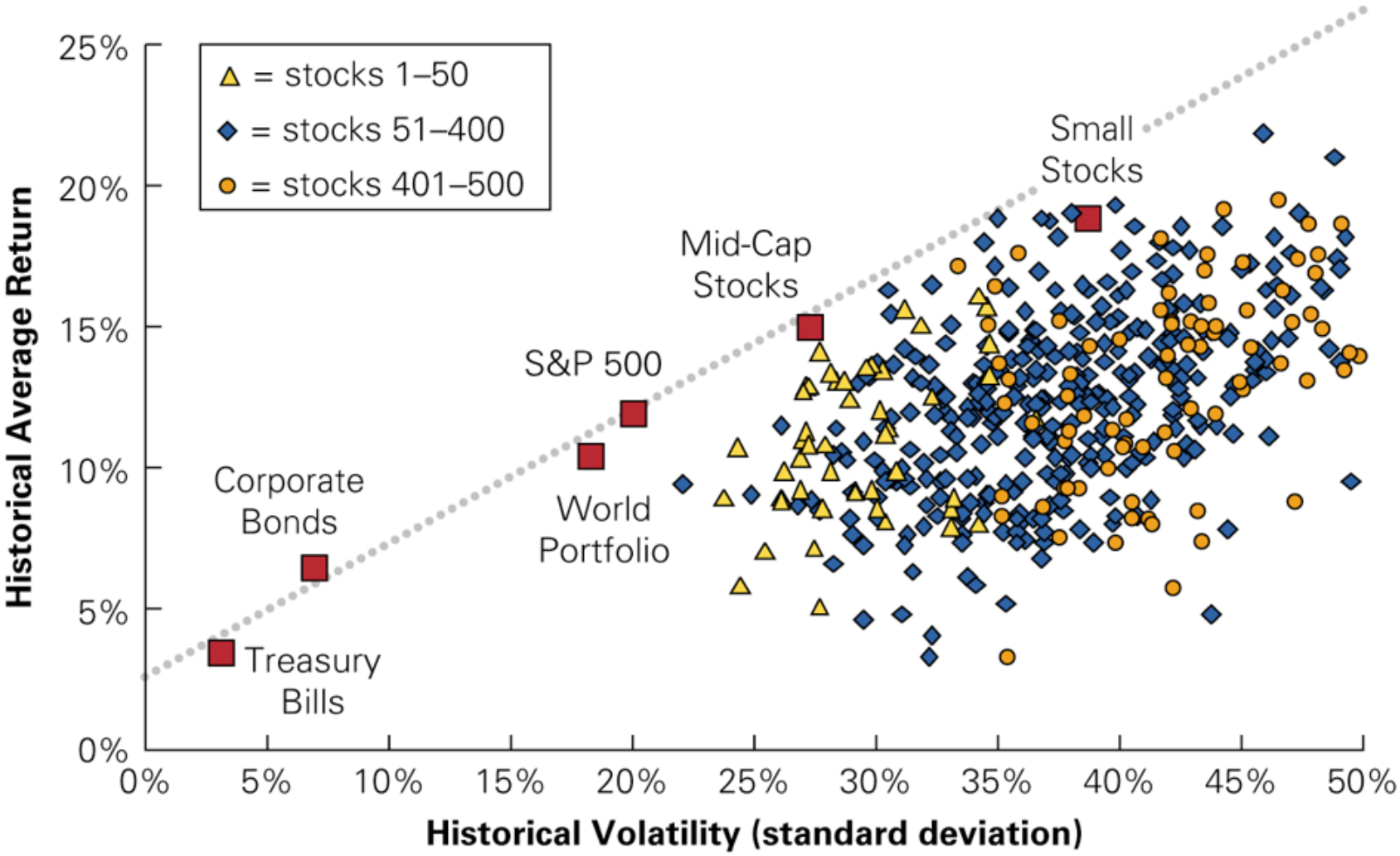
Return/risk of different US portfolios



Source: CRSP, Morgan Stanley Capital International

F. Risk and returns of individual stocks

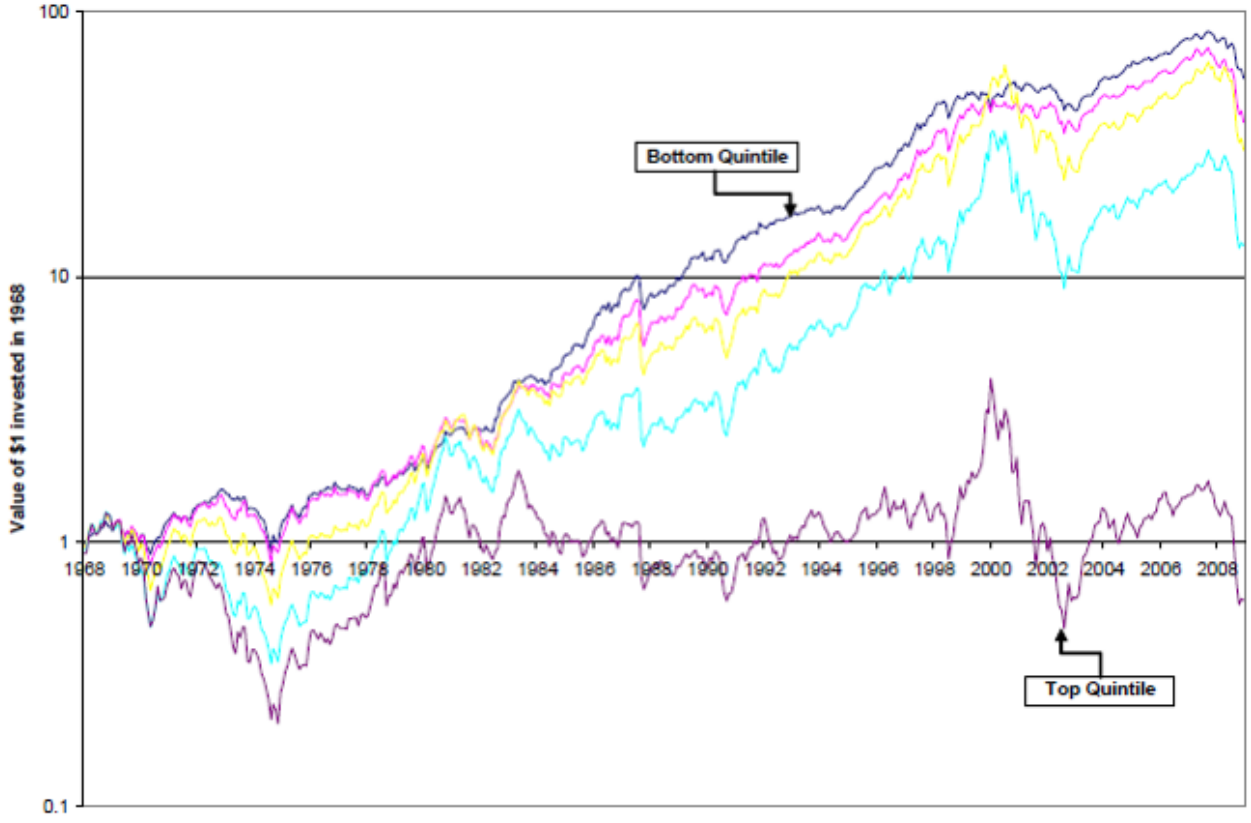
Volatility and realized returns on S&P 500 stocks



Source: CRSP, Morgan Stanley Capital International

What's "new"? The low-risk anomaly

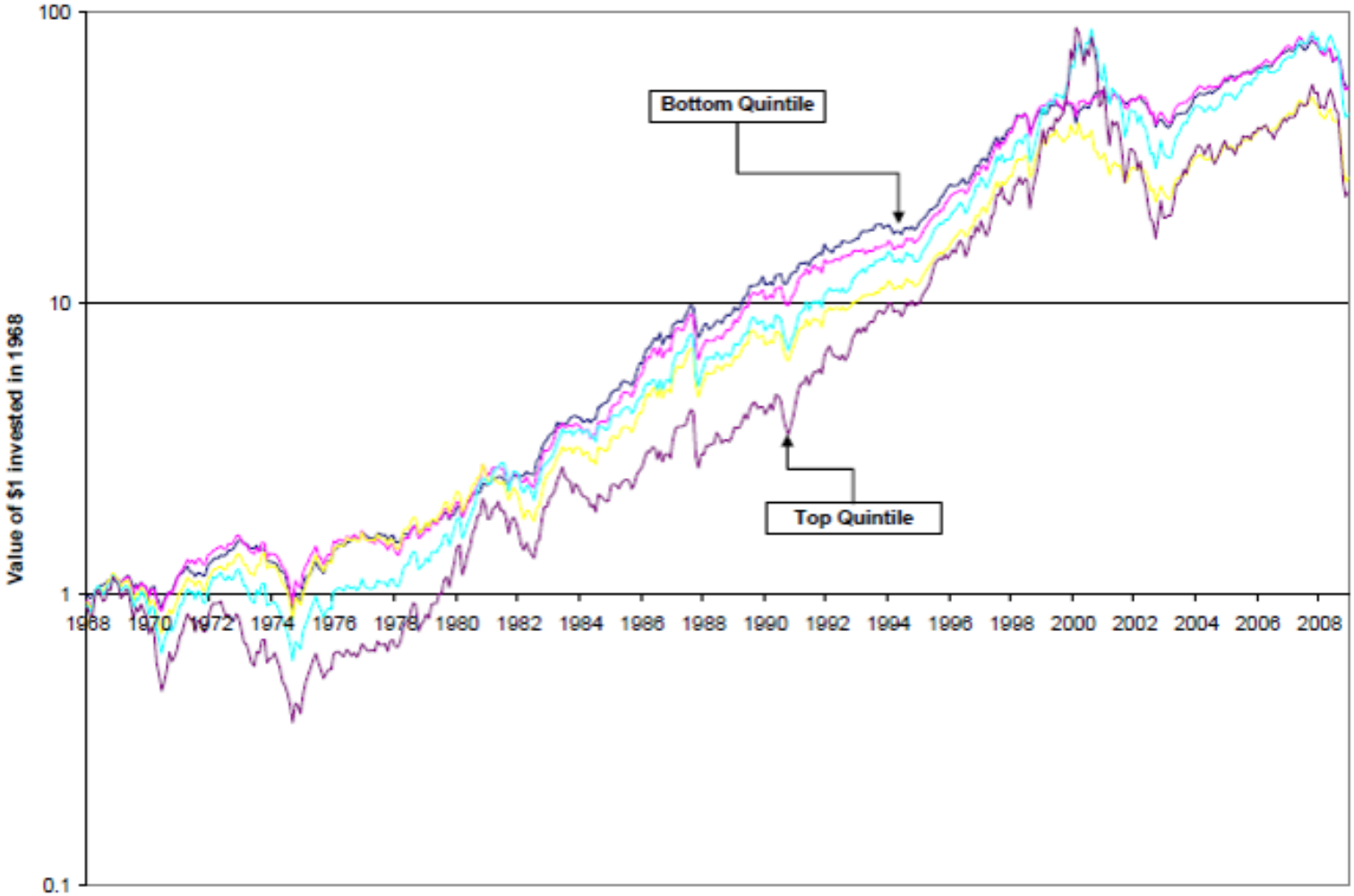
All stocks:



Baker, Bradley and Wurgler (2011).

What's "new"? The low-risk anomaly (con't)

Top 1000 stocks by MktCap



Baker, Bradley and Wurgler (2011).

Quiz

- A stock is worth \$1 at the beginning of the year. At the end of the year and after a reverse stock split of 1:10 and a dividend of \$0.03 (before the split), the stock is worth \$11. What is the return on the investment?
- The value of a stock is halved over six month (without stock split and dividends).
  - What is the return on the investment over the six months?
  - What is the annualized return?
- The value of a stock loses 20% and then gains 20% (without stock split and dividends).
  - What is the average arithmetic return?
  - What is the average geometric return?

Quiz

- Your banker explains to you that stocks always offer a better return than your investment in treasury bills.
  - Is this true?
  - Is it a reason to invest in stocks?
- Can volatility become negative?
- Is the return on a stock with more risk always higher than the return on a stock with less risk?

Critical thinking

- A bond is worth \$97 (for a nominal value of \$100) at the beginning of the year. The bond pays an annual coupon of 4.5%, and is worth \$102 at the end of the year. What is the return on the investment?
- A company's stock price increases by 10% over the year, but the return on equity (ROE) of the company is -30%. Is there a contradiction?